



YOUR TECHNOLOGY & INNOVATION PARTNER

Production Deployment & Operations for AI Agents



PRESENTATION AGENDA

1

SET THE STAGE



PROTOTYPE
VS
PRODUCTION

2

CORE ARCHITECTURE



AGENT STACK
ANATOMY

3

DEPLOYMENT & OPS



CI/CD &
RESILIENCE

4

SECURITY & IMPROVEMENT



GUARDRAILS &
DATA FLYWHEEL

5

Q&A



PROTOTYPE REALITY

Works on **local!**

Serves 1 user



AI agent



PRODUCTION REALITY

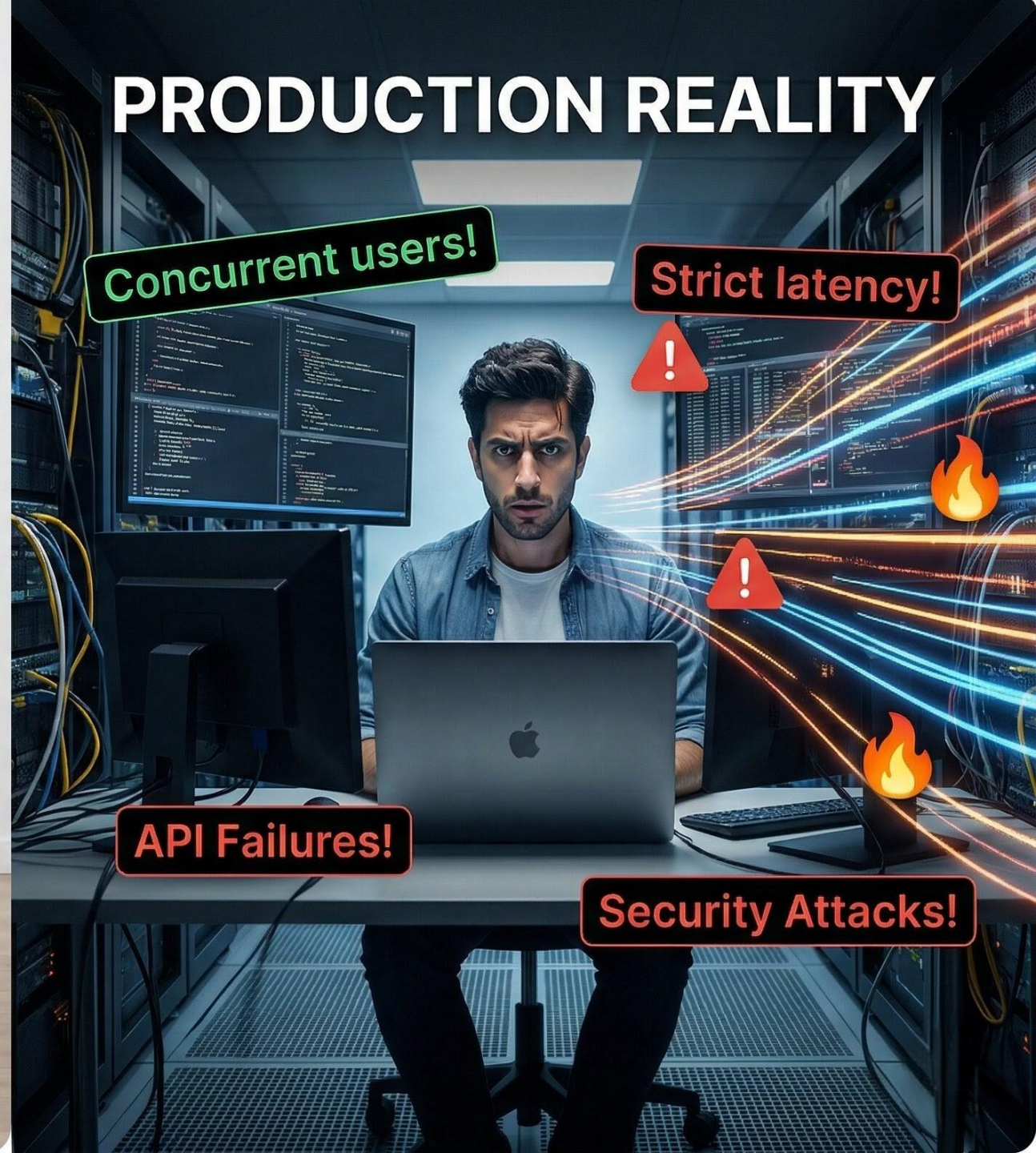
Concurrent users!

Strict latency!



API Failures!

Security Attacks!



ANATOMY OF A PRODUCTION AGENT STACK

LLM ENGINE
(OpenAI, Gemini, vLLM)



AGENT CORE
(ReAct Loop)



KNOWLEDGE BASE / RAG
(Vector DB: Pinecone, Weaviate)



STATE & MEMORY



Web Search
API



Code
Interpreter



External
API Calls

TOOL EXECUTION ENVIRONMENT / SANDBOX

STATE & MEMORY
(Redis, Postgres)

LATENCY & COST OPTIMIZATION (SEMANTIC CACHING)

Incoming stream
of slightly varied
USER QUERIES

USER
QUERIES

SEMANTIC CACHE
similarity search,
cluster visual

CACHE HIT!

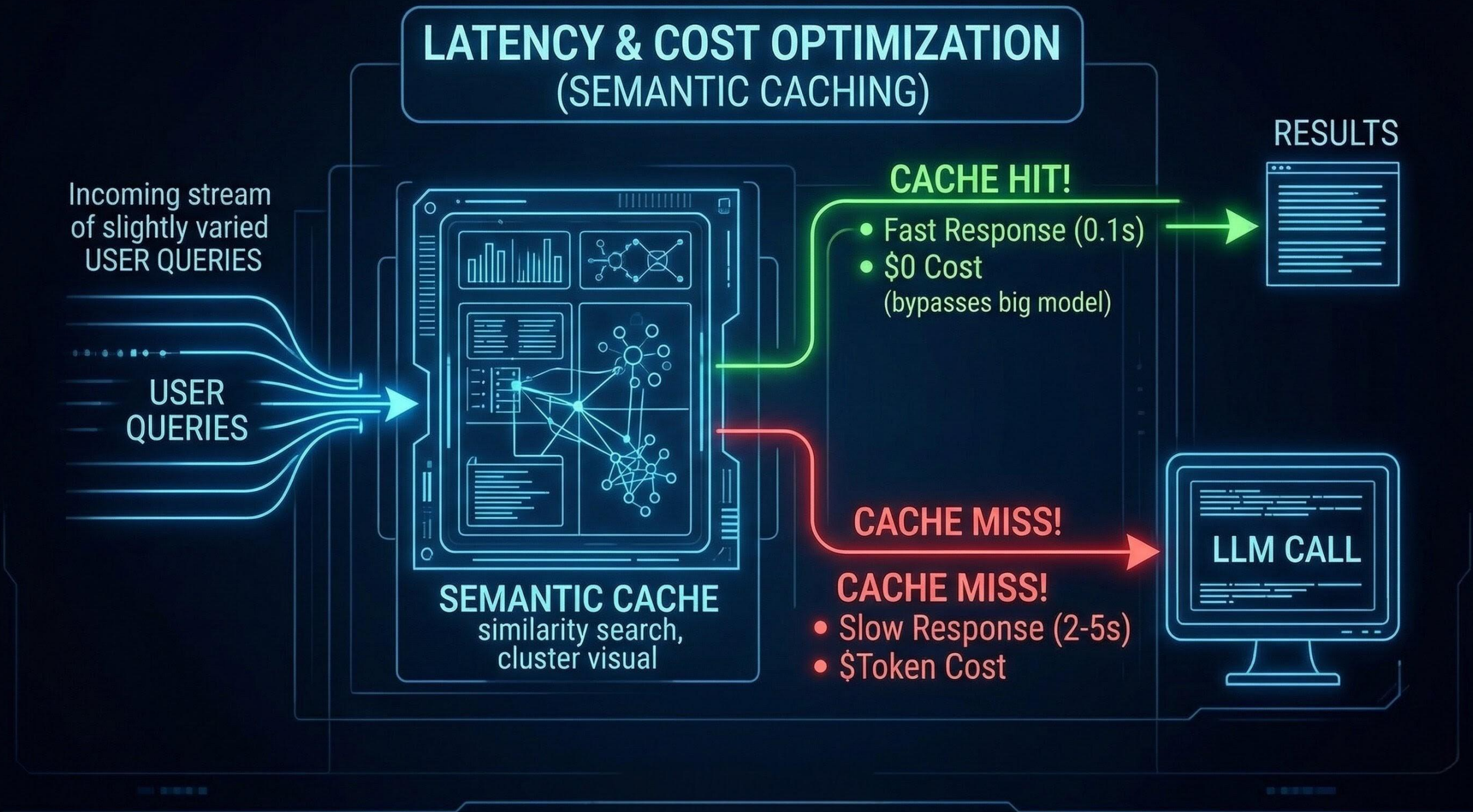
- Fast Response (0.1s)
- \$0 Cost
(bypasses big model)

RESULTS

CACHE MISS!

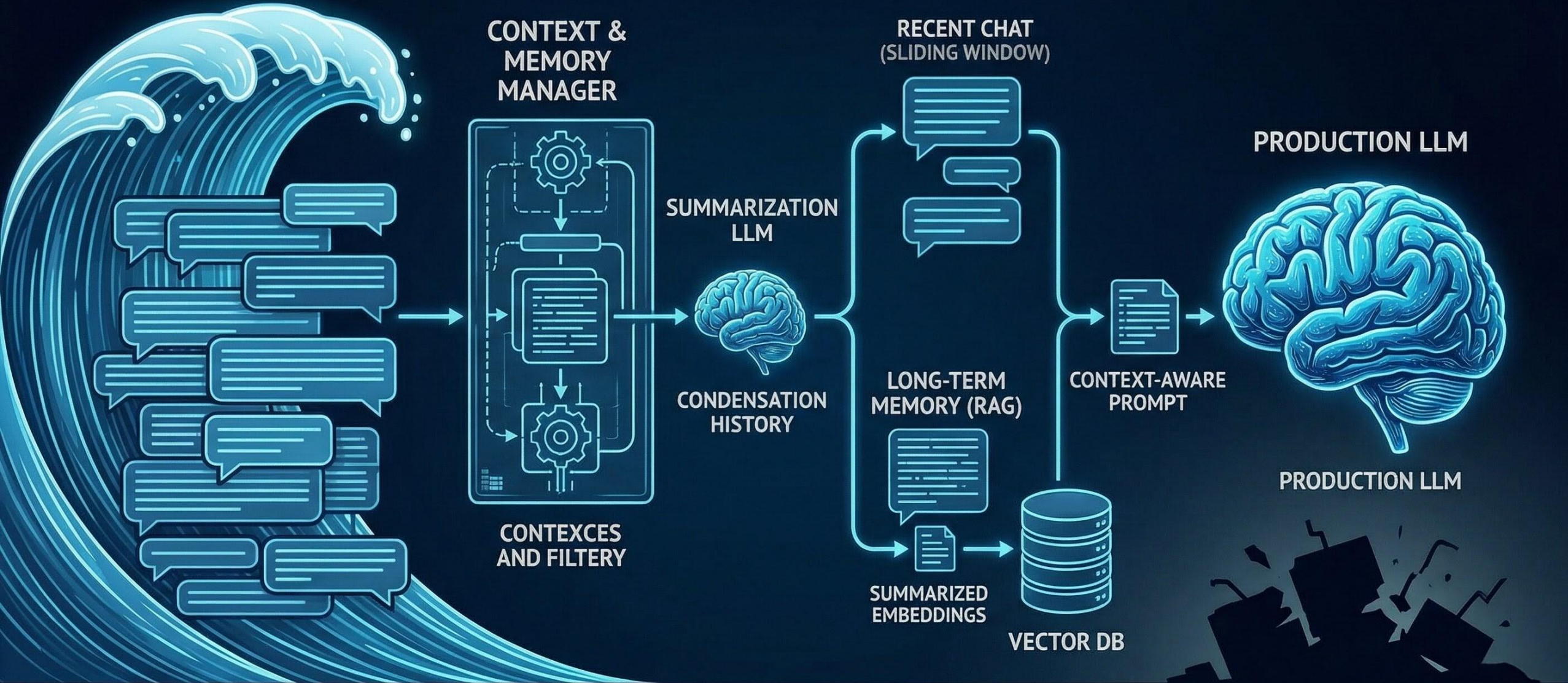
- CACHE MISS!**
- Slow Response (2-5s)
 - \$Token Cost

LLM CALL



MANAGING CONTEXT WINDOW IN PRODUCTION

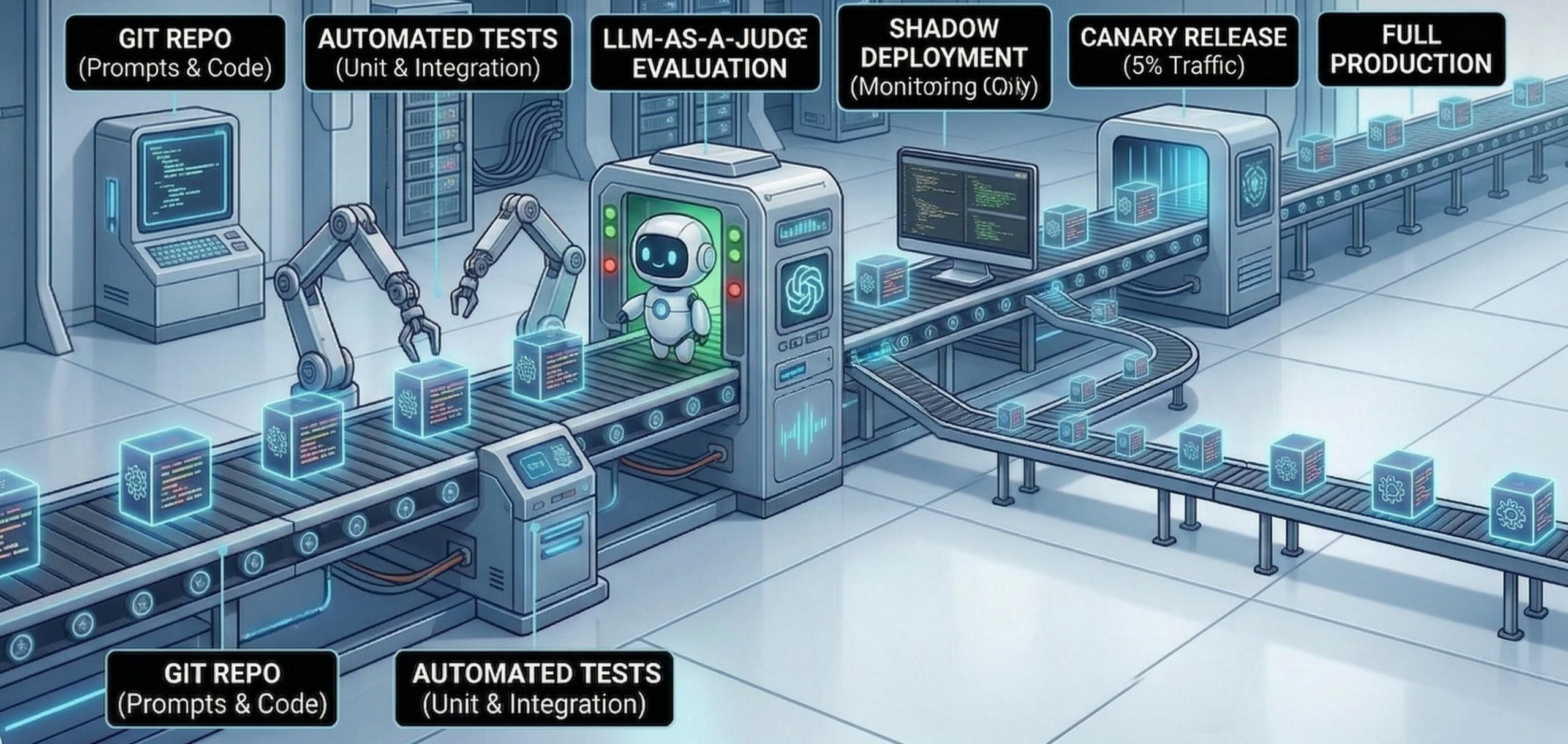
USER CHAT HISTORY



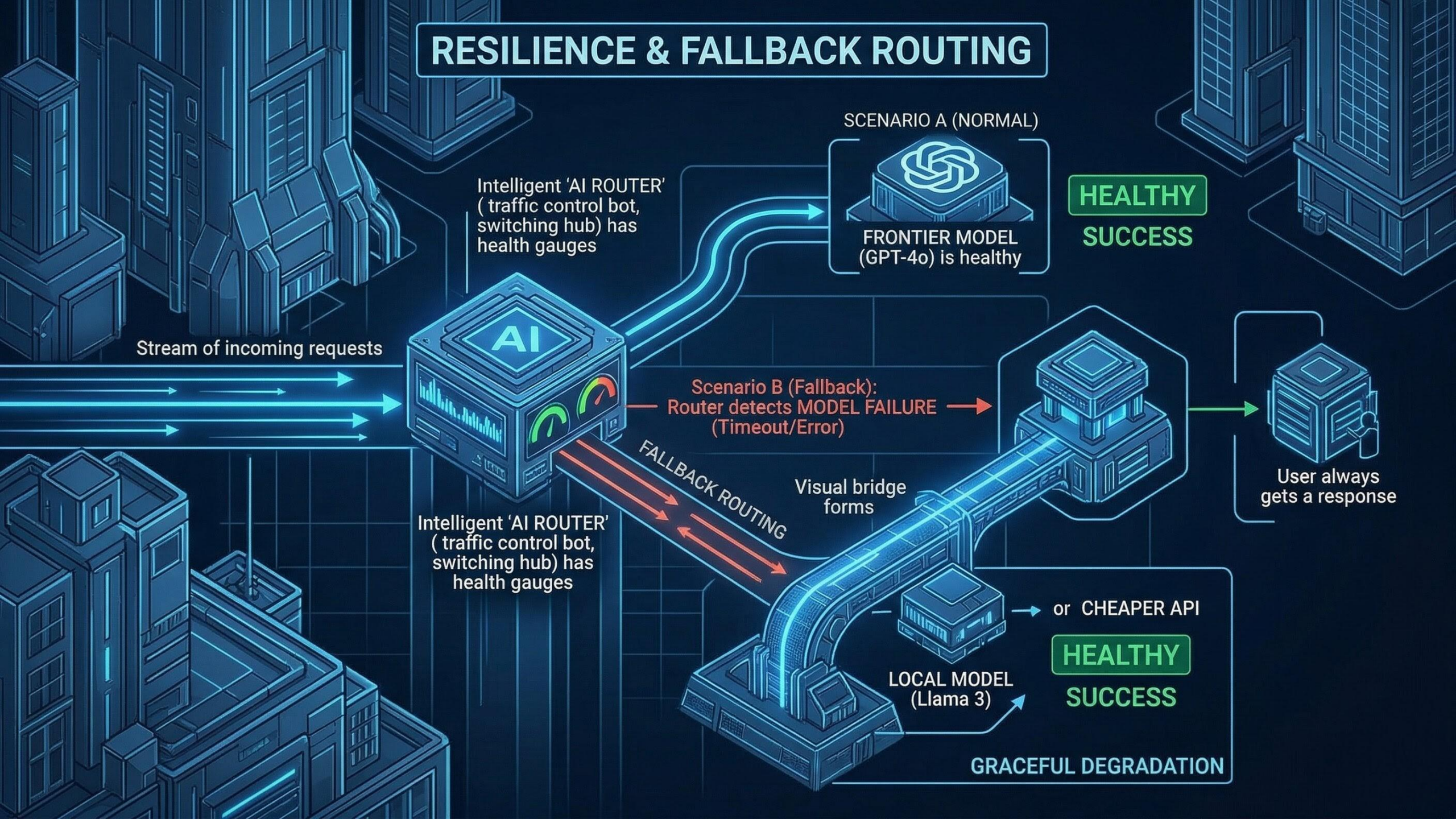
USER CHAT HISTORY

OLD SYSTEM
(OVERFLOW ERROR)

Deployment Strategies & CI/CD



RESILIENCE & FALLBACK ROUTING





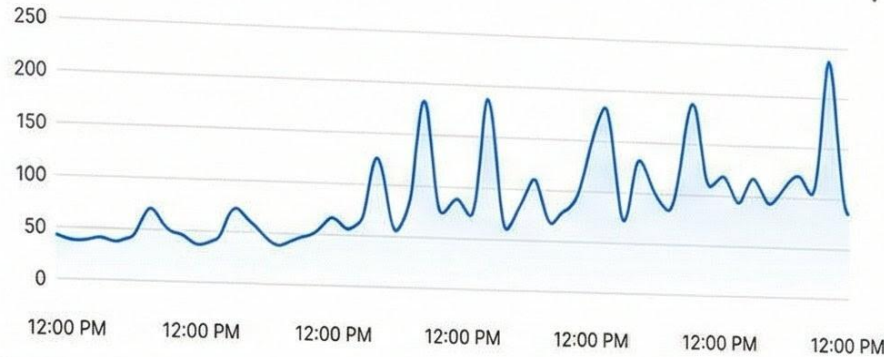
AgentOps - Observability



Static nur linersal

Monitored

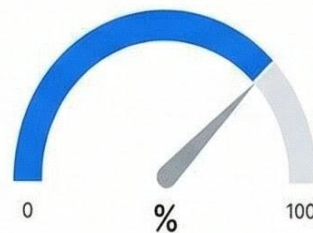
Latency (ms)



Token Usage Cost



Tool Success Rate (%)



Agent Thought Trace

Web Search

SQL API Call

1 Reasoning reasoning Web Search usee is to useet this reasoning.

Reasoning reasoning

2 SQL API Call

Reasoning reasoning Web Search li to cohnite the next SQL API Call shouldn't chang to pling for SQL API is reasoning.

Reasoning-step reasoning

SQL API Call

3 SQL Feam in tannia

SQL API Call

4 Reasoning the reasoning to tine CPM

SQL API Call

Security from Day 1

**USER-INPUT
PROMPT**

INPUT GUARDRAIL
(Prompt Injection/PII Detection)

**AI AGENT
CORE**

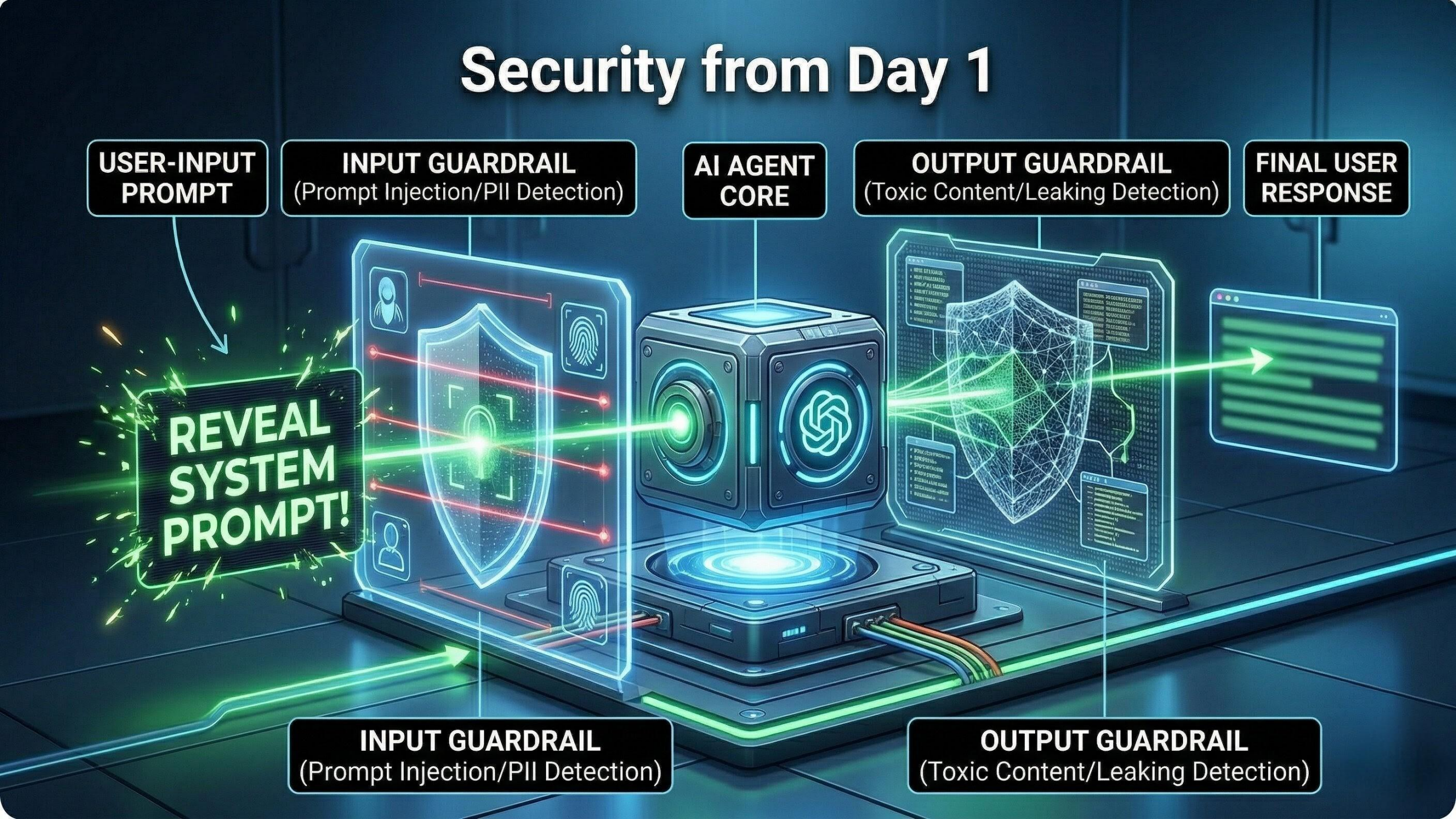
OUTPUT GUARDRAIL
(Toxic Content/Leaking Detection)

**FINAL USER
RESPONSE**

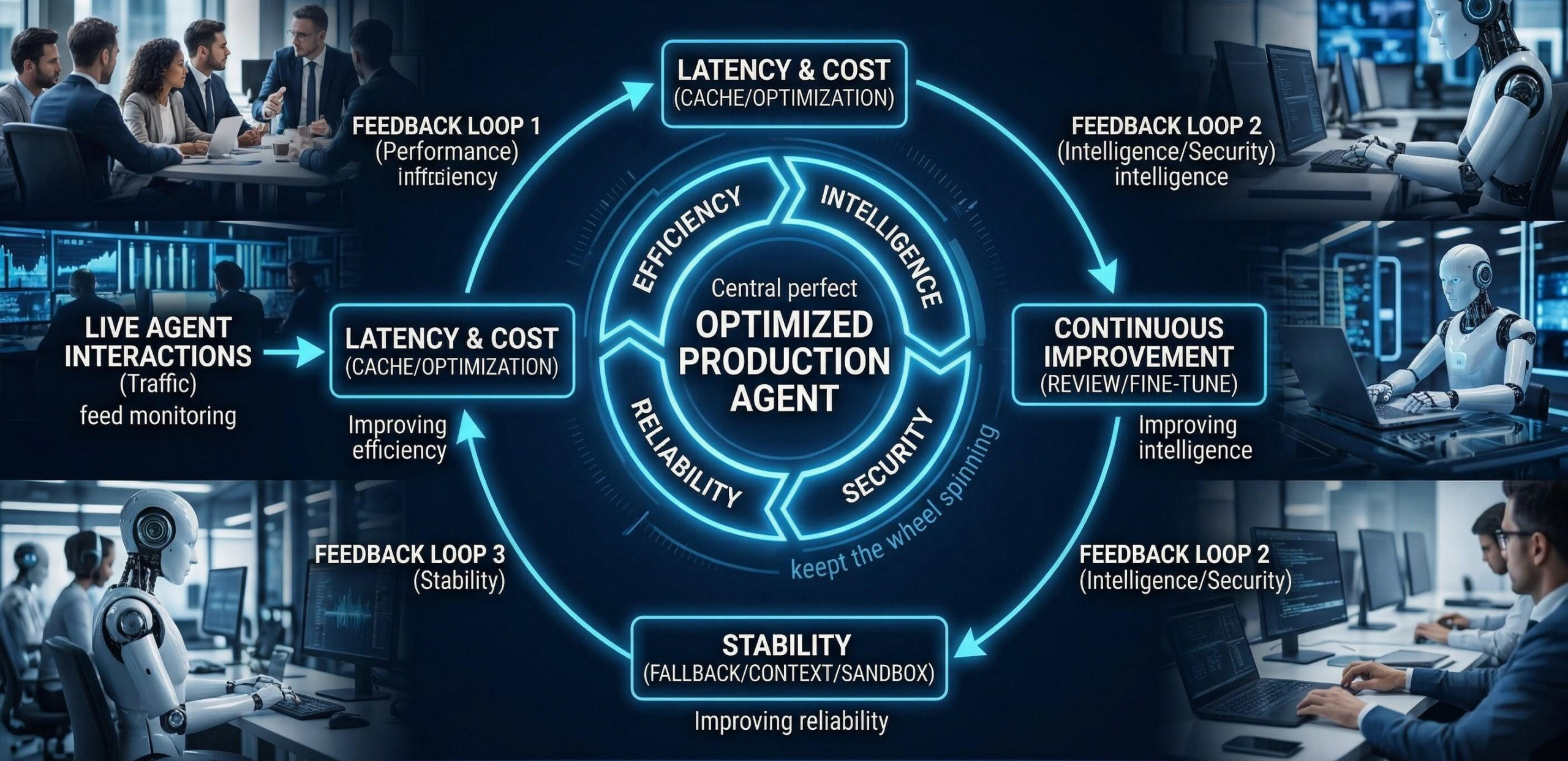
**REVEAL
SYSTEM
PROMPT!**

INPUT GUARDRAIL
(Prompt Injection/PII Detection)

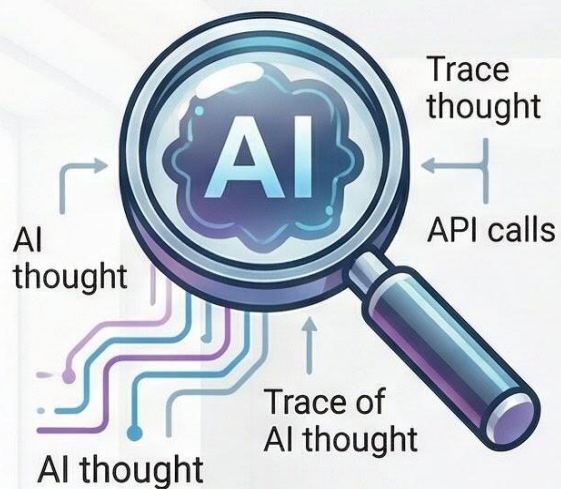
OUTPUT GUARDRAIL
(Toxic Content/Leaking Detection)



CONTINUOUS IMPROVEMENT IN PRODUCTION OPERATIONS



Key Takeaways & Q&A



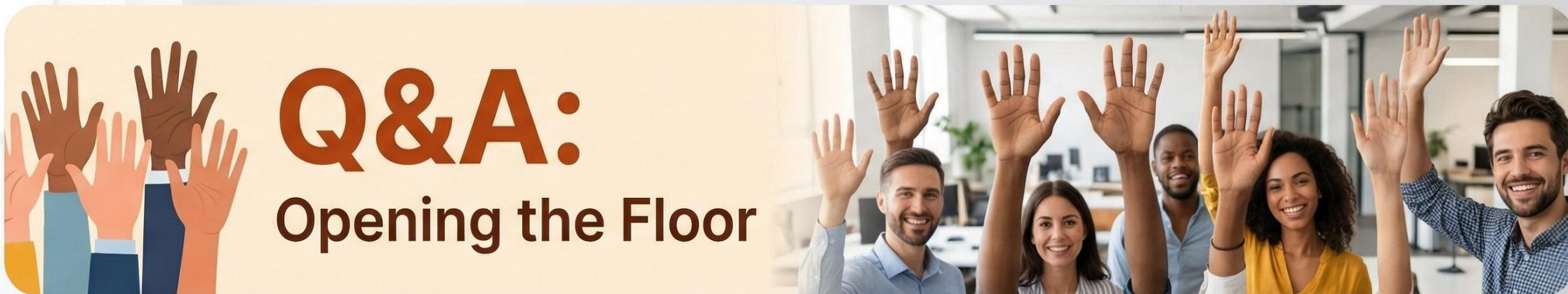
1. AgentOps is critical



2. Expect failure
(Degrade gracefully)



3. Security from Day 1





THANK YOU!

Vietnam: **+84 28 3995 1060**

+84 28 3995 1059

Australia: **+61 411 276 342**

Email: **sales@tmasolutions.com**

North America: **+1 844 224 4188**

Europe: **+33 7 44 77 49 24**

Japan: **+81 50 54 90 98 67**

Website: **www.tmasolutions.com**